



Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin
Ollscoil Átha Cliath | The University of Dublin

Faculty of Science, Technology, Engineering & Mathematics

School of Computer Science & Statistics

Integrated Computer Science Programme
Business and Computing Programme
Management Science and Information Systems Studies Programme
Year 3

Semester 2 2024

Information Management II

Thursday 25 April 2024

Sports Centre

09:30 – 11:30

Prof Yvette Graham

Instructions to Candidates:

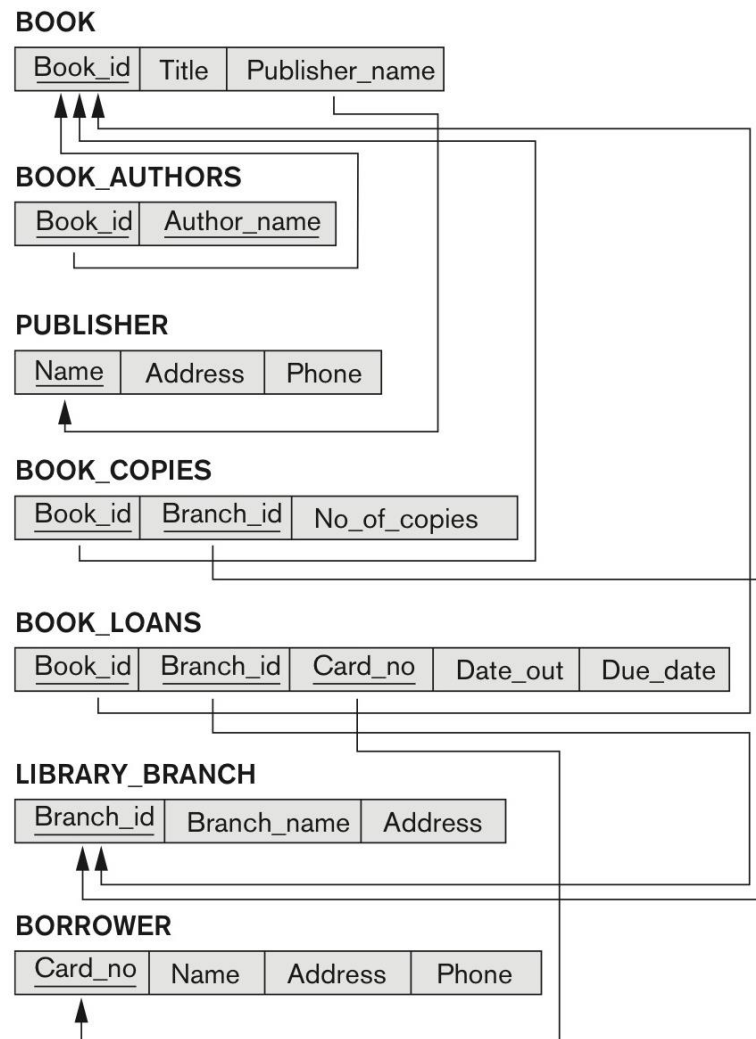
Attempt **three** questions in total. **Question 1 is mandatory.** Answer **any two** questions from **Question 2, Question 3 and Question 4**

All questions carry equal marks. Each question is scored out of a total of 25 marks.

Do not begin the exam until instructed to do so by an invigilator.

Materials permitted: None

Q. 1



A library loan system contains a relational database that uses the above relational database schema that has been produced by mapping from an Entity Relation diagram. Provide a possible Entity Relation diagram that could have been used to produce the relational database schema above. State any assumptions you make.

[25 marks]

Q 2.

COMPANY Relational Database Schema:

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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PROJECT

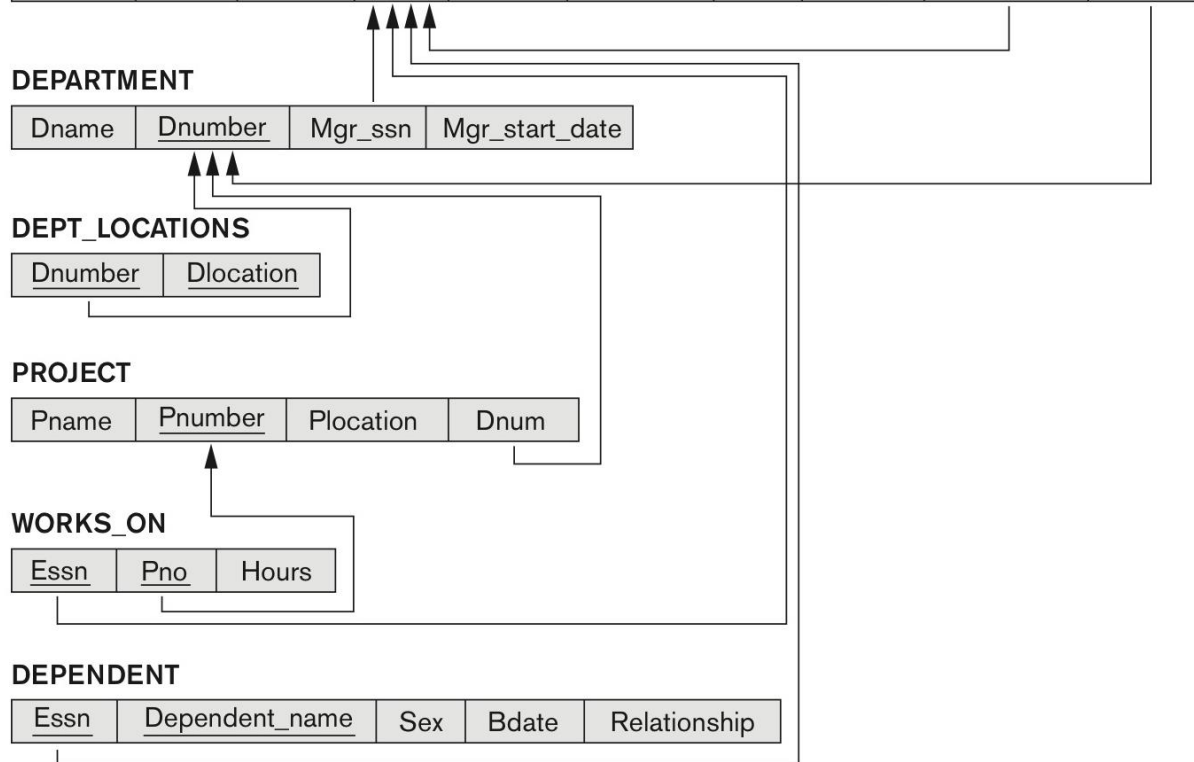
Pname	<u>Pnumber</u>	Plocation	Dnum
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WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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COMPANY Relational Database State:

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Suppose that each of the following Update operations is applied directly to the above COMPANY relational database schema and database state. Discuss all integrity constraints violated by each operation (a) to (e), if any, and the different ways of enforcing these constraints.

- (a) Insert <'Sophia', 'M', 'Wood', '973442298', '1974-05-21', '23 S Lamar Blvd. Rd, Austin, TX', 'F', 62000, '222445555', 5> into EMPLOYEE. [5 marks]
- (b) Insert <'Information Technology', 2, '987987987', '2007-10-01'> into DEPARTMENT. [5 marks]
- (c) Insert <'777624972', 15, '40.0'> into WORKS_ON. [5 marks]
- (d) Delete the DEPARTMENT tuples with Dnumber = 5. [5 marks]
- (e) Modify the Pnumber attribute of the PROJECT tuple with Pnumber = 30 to 40 [5 marks]

Q 3.

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
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WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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Consider the above relational database schema. Suppose that all the relations were created by (and hence are owned by) user X, who wants to grant the following privileges to user accounts A, B, C, D, and E. Write SQL statements to grant these privileges:

- Account A can retrieve or modify any relation except DEPENDENT and can grant any of these privileges to other users.
[5 marks]
- Account B can retrieve all the attributes of EMPLOYEE and DEPARTMENT except for SALARY, MGRSSN, and MGRSTARTDATE.
[5 marks]
- Account C can retrieve or modify WORKS_ON but can only retrieve the FNAME, MINIT, LNAME, SSN attributes of EMPLOYEE and the PNAME, PNUMBER attributes of PROJECT.
[5 marks]
- Account D can retrieve any attribute of EMPLOYEE or DEPENDENT and can modify DEPENDENT.
[5 marks]
- Account E can retrieve any attribute of EMPLOYEE but only for EMPLOYEE tuples that have DNO = 3.
[5 marks]

Q 4.

- i) What is GDPR? Include in your answer what jurisdiction GDPR applies to and what legislation it now replaces.
[8 marks]
- ii) List the GDPR principles.
[8 marks]
- iii) What is personal data?
[9 marks]